

VIP C2 PROGRAMME — FULL STACK DEVELOPMENT WITH
MERN

Data Flow Diagrams & User Stories

Online Complaint Registration System — MERN Stack Complaint
Management Platform

Phase: Brainstorming & Ideation Phase

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Data Flow Diagrams & User Stories

1. Introduction

This document presents the Data Flow Diagrams (DFDs) and formal User Stories for the Online Complaint Registration System [cite: 940]. The Data Flow Diagrams model the system at multiple abstraction levels, including the Context Diagram (Level 0) and Functional Decomposition Diagrams (Level 1), illustrating the movement of information between users, processes, external systems, and data stores [cite: 941, 942]. The User Stories define the functional requirements from the perspectives of Citizens, Service Agents, and Administrators [cite: 925]. These stories capture system behavior, user expectations, acceptance criteria, sprint allocation, and development priorities [cite: 926].

2. Level 0 – Context Diagram

The Level 0 diagram visualizes the structural core system as a black box interacting with external stakeholder entities and surrounding cloud APIs [cite: 941, 946].

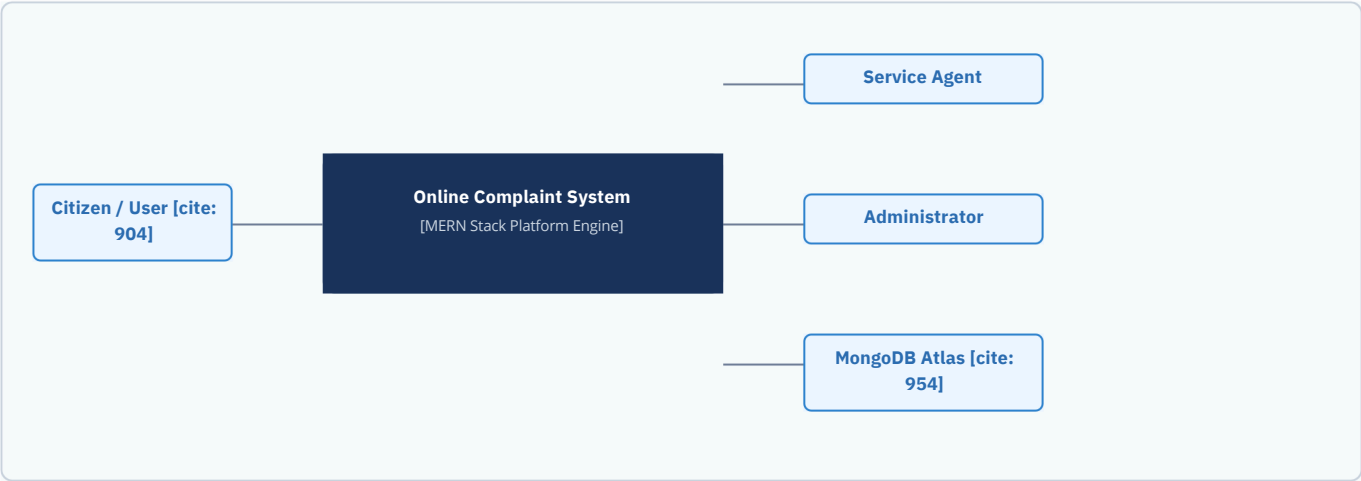


FIGURE 6.1 – LEVEL 0: CONTEXT DIAGRAM (ONLINE COMPLAINT REGISTRATION SYSTEM) [CITE: 946]

3. Level 1 – Authentication Sub-Process

This functional decomposition models the operational parameters governing authorization, validation middleware loops, and password token persistence [cite: 948].

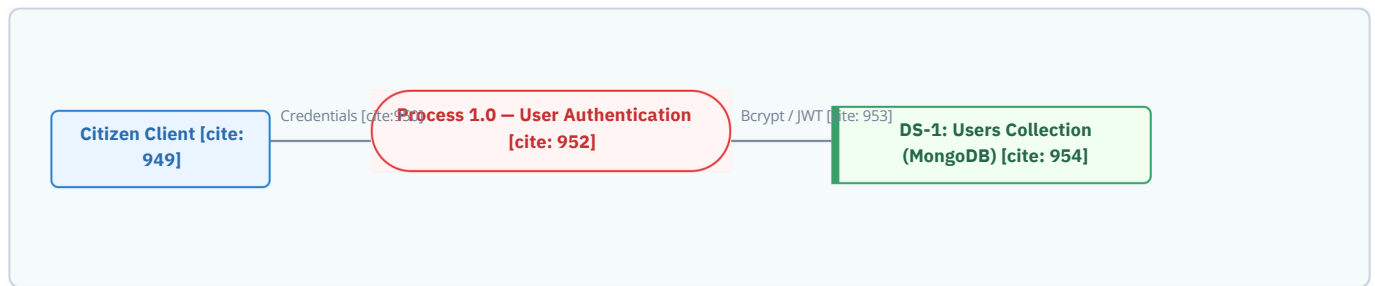


FIGURE 6.2 – LEVEL 1: AUTHENTICATION PROCESS DFD [CITE: 956]

4. Level 1 – Complaint Registration & Tracking Sub-Process

Decompose the core transaction lifecycle initiated by citizens from point of initial structural record log down to resolution feedback parameters [cite: 958].

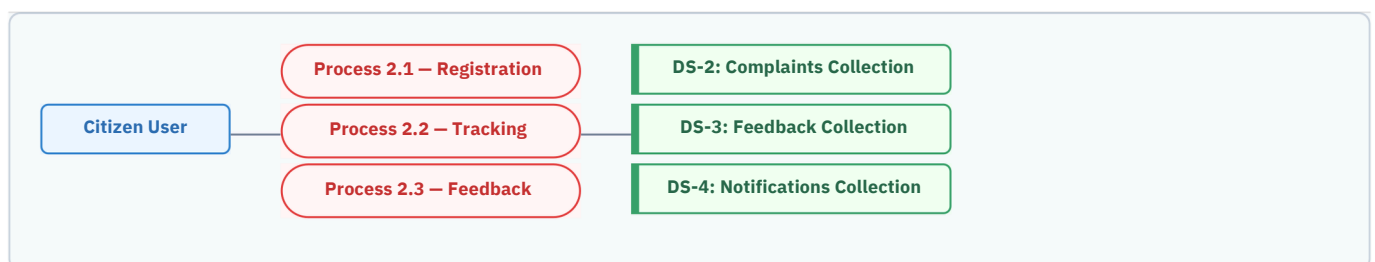


FIGURE 6.3 – LEVEL 1: COMPLAINT REGISTRATION & TRACKING DFD [CITE: 962]

5. Level 1 – Agent Operations & Communication Sub-Process

Illustrate the operational data exchanges and full WebSocket connection handshakes active during agent mediation [cite: 958].

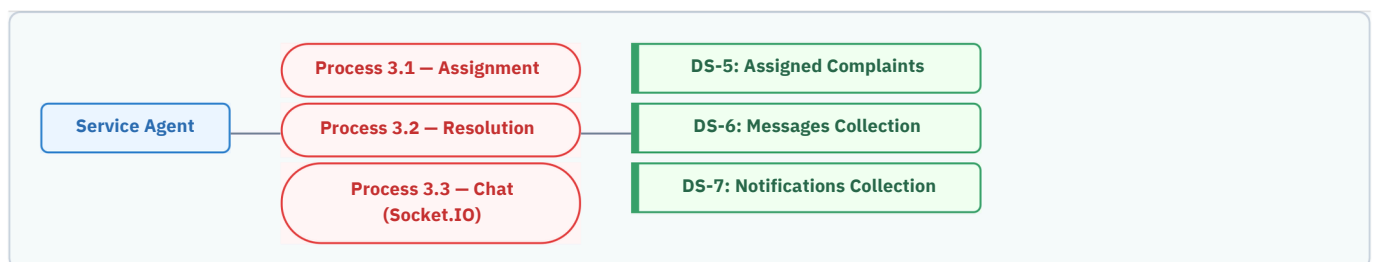


FIGURE 6.4 – LEVEL 1: AGENT OPERATIONS & COMMUNICATION DFD

6. User Stories

USN [cite: 968]	User Type [cite: 971]	User Story [cite: 969]	Acceptance Criteria [cite: 972]	Points [cite: 970]	Sprint [cite: 970]
USN-1	Citizen	As a citizen, I can register using my name, email, and password [cite: 964].	Account created successfully; redirected to main dashboard [cite: 964].	2 [cite: 964]	Sprint 1 [cite: 964]
USN-2	Citizen	As a citizen, I can log in securely using my registered credentials [cite: 964].	JWT authorization token issued in a secure HTTP-only cookie [cite: 964].	2 [cite: 964]	Sprint 1 [cite: 964]
USN-3	Citizen	As a citizen, I can submit complaints with category and issue details.	Complaint record stored in database; unique complaint ID generated.	3	Sprint 2
USN-4	Citizen	As a citizen, I can track complaint status throughout the resolution process.	Status lifecycle stages are displayed and updated in real time [cite: 964].	3 [cite: 964]	Sprint 2 [cite: 964]
USN-5	Citizen	As a citizen, I can communicate with assigned agents through real-time chat.	Messages delivered instantly via Socket.IO connections [cite: 964].	5 [cite: 964]	Sprint 3 [cite: 964]
USN-6	Citizen	As a citizen, I can provide ratings and feedback after complaint resolution [cite: 964].	Feedback document recorded and visible to administrators [cite: 964].	2 [cite: 964]	Sprint 3 [cite: 964]
USN-7	Service Agent	As a service agent, I can view complaints assigned to me [cite: 964].	Assigned case profiles populate table inside agent workspace panel.	3	Sprint 2
USN-8	Service Agent	As a service agent, I can update complaint status and provide resolution remarks [cite: 964].	Status transitions updated cleanly across all dashboard modules [cite: 964].	3	Sprint 2
USN-9	Service Agent	As a service agent, I can communicate with citizens using Socket.IO chat.	Real-time chat messaging functions reliably; socket hooks operational.	3	Sprint 3
USN-10	Administrator	As an administrator, I can approve or reject service agent registrations.	Pending application validation interface workflow updates state properly.	2	Sprint 1
USN-11	Administrator	As an administrator, I can monitor complaints and manage users.	Central tracker provides complete system metrics and data views [cite: 964].	3	Sprint 3
USN-12	Administrator	As an administrator, I can view analytics and performance reports [cite: 964].	Admin dashboard graphs show resolution trends and metric summaries [cite: 964].	3 [cite: 964]	Sprint 3 [cite: 964]

7. Conclusion

The Data Flow Diagrams provide a structured representation of how information moves across the Online Complaint Registration System, illustrating interactions between citizens, service agents, administrators, and supporting services. The User Stories define the functional requirements from each stakeholder perspective and establish a clear roadmap for Agile development and sprint planning. Together, they guarantee seamless traceability throughout system implementation.

Additional Resources Required

- Level 2 Functional Decomposition schemas mapping granular socket event payload handshakes.
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